Predicting Heart Disease from Survey Answers: the Codebook Matters More than the Regularizer

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***Abstract*—We predict coronary heart disease (MICHD) for 328,135 respondents of the 2015 BRFSS telephone survey from their 321 self-reported answers, using logistic regression trained by gradient descent and scored by F1. Cross-validation shows that the model cannot overfit ($N/D \approx 2{,}100$), so L2 regularization only hurts, and the decisive choices are the decision threshold and how answers are encoded. BRFSS stores "don't know" and "refused" as per-variable numeric codes that a linear model reads as extreme values; recoding them to missing with the official codebook is the single representation change that helps ($+0.013$ CV F1). Our final model reaches a 5-fold CV F1 of 0.425. We also report three negative results: one-hot encoding, squared features and class rebalancing.**

I. Introduction

The task is binary classification with a strong imbalance: 8.83% of training rows are positive. Accuracy is therefore useless (predicting "healthy" for everyone scores 91%), and the competition metric is the F1 score of the positive class. Predicting *everyone* positive already gives $\text{F1} = 2p/(1+p) = 0.162$, which is the floor any model has to beat. Every feature is a questionnaire answer: there are no lab values, and 44.7% of all cells are missing, mostly because of the survey's skip logic.

Only NumPy is allowed, so we stay with linear models and spend the effort on validation and on how the answers are represented. All full-data results are reproduced by `run.py` and `ablation.py` in our repository.

II. Preprocessing

Every statistic below is fitted inside each training fold only.

1. Sentinel codes $\rightarrow$ missing. BRFSS codes a non-answer just above each variable's valid range, and the code differs per variable: 7/9 for `GENHLTH` (scale 1–5), 77/99 for the 0–30 day counts (where 7 and 9 are real answers), 7777/9999 for weight in pounds, 3 for `_AGE65YR`, 14 for `_AGEG5YR`. Left in place, these codes are read as magnitudes. Fitting logistic regression on `GENHLTH` alone gives a slope of $+0.80$ per standard deviation, i.e. the odds are multiplied by 2.2 per step. A refusal (9) standardizes to $+5.8\sigma$, 3.6σ beyond the worst real answer, where the model predicts an 88% risk against an observed 14%. The observed risk rises from 1 to 5 and then *falls* at 7 and 9, a shape no single coefficient can fit. We took the codes of 242 of the 321 columns from the official CDC 2015 SAS format files [1] (`brfss_codebook.py`). We first tried to infer them from the data ("a code is a sentinel if the value below it never occurs"), but this rule cannot find `_AGE65YR`$=3$ or `_AGEG5YR`$=14$, which follow their real values without a gap, and a cruder "7 and 9 are sentinels" rule would have erased real day counts.

2. Column filtering. We drop columns that are more than 50% missing after step 1, constant columns, and 28 survey administration columns (record numbers, sampling weights, interview dates, telephone screening, and the state FIPS code, whose numeric order is meaningless). 152 columns remain.

3. Standardization, then missing $\rightarrow 0$. Each column is centred and scaled with the mean and standard deviation of its *observed* values; missing entries are then set to 0. After centring, 0 is the column mean, so a missing answer adds nothing to the score $w^\top z$. Any other value is a claim about where the respondent sits on the health axis. Imputing a raw 0 before scaling would be wrong, because 0 lies outside the valid range of most answers.

4. Two counts and an intercept. Recoding a refusal to missing merges it with "never asked". To keep the distinction we add, per row, the standardized number of refused answers and of questions never asked. Rows with at least three refusals are 10.4% positive, compared with 4.7% for rows with none. Together with the intercept this gives $D = 155$.

III. Models and Validation

Methods. We implemented the six required methods (MSE by gradient descent and SGD, least squares and ridge by normal equations, logistic and L2-regularized logistic regression by gradient descent), all checked against the course's reference values. Least squares on $\{0,1\}$ labels serves as a baseline. The final model is logistic regression trained by full-batch gradient descent from $w = 0$.

Step size and iterations. After 500 steps the training loss is 0.2292 for $\gamma = 0.1$ and 0.2258 for $\gamma = 1$, and gradient descent diverges for $\gamma \geq 5$ (loss 0.697, above the 0.693 of $w = 0$). We use $\gamma = 1$ and 500 iterations. On a held-out 20%, 4000 iterations change validation F1 by only $+0.001$, and validation loss is lowest at 500.

Cross-validation. We use 5 folds (seed 1). Preprocessing is refitted inside every training fold, so no validation row influences it; all five folds keep exactly the same columns as a fit on the full data. The decision threshold plays no part in training, so we cache each fold's validation probabilities and sweep 56 thresholds from 0.05 to 0.60 in steps of 0.01 at no extra cost. We select the $(\lambda, \text{threshold})$ pair with the best mean validation F1, then refit preprocessing and weights on all training rows. The CV estimate is slightly pessimistic, because each fold trains on 80% of the data. It is also a good

TABLE I

ABLATION ON THE FULL TRAINING SET, 5-FOLD CV, MEAN ± STD OVER FOLDS. EACH LOGISTIC ROW ADDS ONE CHANGE TO THE ROW ABOVE. $\bar{p}$ IS THE MEAN PREDICTED PROBABILITY; THE BASE RATE IS 0.0883.

Configuration	D	CV F1	thr.	$\bar{p}$
Predict all positive	–	0.162	–	–
Least squares, final features	155	0.416 ± 0.004	0.21	–
Logistic, γ=0.1, λ=1, all columns	173	0.380 ± 0.004	0.50	0.455
+ $\lambda = 0$	173	0.408 ± 0.003	0.21	0.098
+ $\gamma = 1$	173	0.412 ± 0.003	0.20	0.088
+ drop survey metadata	154	0.412 ± 0.003	0.20	0.088
+ sentinels → missing	153	0.425 ± 0.005	0.20	0.088
+ refused / never-asked counts	155	$\mathbf{0.425 \pm 0.005}$	0.20	0.088
Final model, threshold 0.5	155	0.226 ± 0.005	0.50	0.088

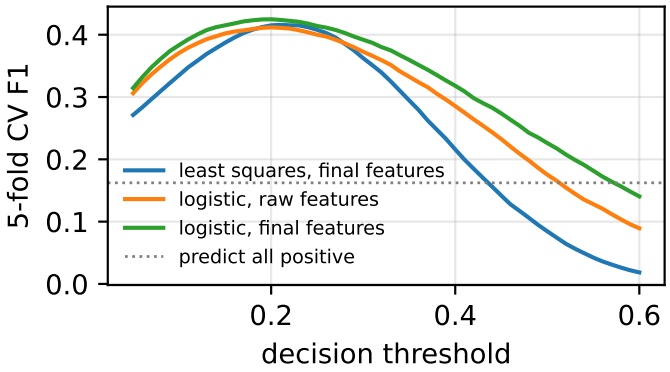


Fig. 1. 5-fold CV F1 against the decision threshold (full training set).

predictor of the leaderboard: an earlier configuration scored 0.393 in CV (on a 10% subsample) and 0.403 on AIcrowd.

IV. RESULTS

Table I adds one change at a time, on the full training set with the same five folds. Figure 1 shows F1 as a function of the threshold.

The threshold is the largest single lever. The final model scores 0.226 at the default threshold of 0.5 and 0.425 at 0.20. For a calibrated classifier, the F1-optimal threshold is half the achievable F1 [2], here $0.425/2 \approx 0.21$, which is close to the 0.20 we select.

Regularization does not help. With $N/D \approx 2{,}100$ there is almost no variance for L2 to control, and the CV and leaderboard scores agree, so there is no overfitting to fix. Removing $\lambda = 1$ is worth $+0.028$ (Table I). Shrinking w, intercept included, pulls every probability towards $\sigma(0) = 0.5$: the regularized model has $\bar{p} = 0.455$ and its best threshold is 0.5. This explains why our earlier submission needed a threshold of 0.4. At $\lambda = 0$, $\bar{p}$ equals the base rate, so the model is calibrated. The grid in `run.py` gives CV F1 0.4248, 0.4252 and 0.4208 for $\lambda = 0$, 0.001 and 0.01. The selection rule picks $\lambda = 0.001$, but its lead is ten times smaller than the fold standard deviation, so in practice $\lambda = 0$ and $\lambda = 0.001$ are equivalent. Larger values could not be tested at $\gamma = 1$: each step multiplies w by $(1 - 2\gamma\lambda)$, so $\lambda = 1$ gives the update $w \leftarrow -w - \nabla L$, and $\lambda = 0.1$ already oscillates (CV F1 0.16). At $\gamma = 0.1$ on a 10% subsample, F1 fell steadily from 0.402 ($\lambda = 0$) to 0.393 ($\lambda = 0.1$) and 0.341 ($\lambda = 5$). Increasing γ from 0.1 to 1 adds $+0.004$, because 500 steps of $\gamma = 0.1$ have not converged ($\bar{p} = 0.098$).

Sentinel recoding is the representation change that helps. Dropping the metadata columns is neutral, which we expected: it is a correctness fix, since record numbers and sampling weights carry no information about the respondent. Recoding the sentinels adds $+0.013$, about three times the fold standard deviation. How the codes are neutralized matters. Excluding them from the mean and standard deviation while still passing the raw value is catastrophic (CV F1 0.22 on a 10% subsample, against 0.40 when they are left untouched): the codes had been inflating σ and so compressing themselves, and without them they land even further out. Adding one "refused" indicator per column (264 columns, on the subsample) gains nothing, and neither do the two counts ($+0.00002$). The association between refusals and the outcome is real, but it is already explained by covariates such as age and general health.

Negative results (10% subsample). *(i) One-hot encoding* the 97 kept columns that are categorical codes with 3–10 levels (772–1,062 columns) *lowered* CV F1 to 0.20–0.27, even after 3000 iterations. The cause is conditioning, not overfitting: a standardized dummy for a level held by 0.1% of rows takes the values -0.03 and $+31.6$, and gradient descent with one step size cannot converge along such directions. The best threshold sat at the bottom of the sweep and $\bar{p}$ far below the base rate. *(ii) Squared features* failed for the same reason (0.32–0.37). *(iii) Undersampling negatives* to 1:1, 2:1, 4:1 and 8:1 gave 0.355, 0.374, 0.392 and 0.402, against 0.402 without it, while the best threshold moved from 0.70 to 0.24. Rebalancing a calibrated model only shifts its optimal threshold [3].

V. DISCUSSION

Our final model ($\lambda = 0.001$, threshold 0.20) reaches a 5-fold CV F1 of 0.425. Its submission (`run.py`) scores 0.4xx on AIcrowd, up from 0.403 for our previous best ($\lambda = 1$, threshold 0.4, no sentinel recoding). Since there is no overfitting, the remaining gap to the best teams (≈ 0.45) has to come from representation, not regularization. The clearest candidate is one-hot encoding restricted to levels with at least 1% support, which keeps the problem well conditioned, or a second-order optimizer that is insensitive to conditioning. Two limitations apply. Both the features and the label are self-reported, so the model predicts a *reported* diagnosis. Hyperparameters were also chosen by CV with the leaderboard as the only external check, so we kept the number of submissions small.

REFERENCES

[1] Centers for Disease Control and Prevention, "Behavioral Risk Factor Surveillance System, 2015 codebook and SAS format files (`Formas15.sas`, `Format15.sas`)," https://www.cdc.gov/brfss/annual_data/annual_2015.html.

[2] Z. C. Lipton, C. Elkan, and B. Narayanaswamy, "Optimal thresholding of classifiers to maximize F1 measure," in *ECML PKDD*, 2014, pp. 225–239.

[3] C. Elkan, "The foundations of cost-sensitive learning," in *IJCAI*, 2001, pp. 973–978.